# I. Context

As our DDF increases in amount and our Department gets more people, it is important to have some guidelines on how funds in the DDF can be spent (or not spent). The current source of the DDF is from WILP courses (hereinafter referred to as W-DDF). The other sources, hereinafter collectively referred to as C-DDF, come from the overhead of external projects that the department faculty members bring. C-DDF also has contributions from consulting, and APPCAIR. There is a DDF scrutiny committee who will review and approve a DDF application.

The proposal described below is valid for one year and can be subject to a periodic review after a year.

# II. DDF Scrutiny Committee

The committee overseeing the DDF usage comprises of the DRC as permanent members. HOD-CSIS will be a part of the committee by default. The committee will include an Assistant Professor, selected on a rotation basis. The responsibilities are:

i. Review each application for presenting a paper in a conference to decide the extent of support to be provided based on the corpus available, the publication venue and the nature of publication. The decision will be based on the policy defined in **Section III**. The committee will attempt to make a fair distribution among the applicants, should the need arise. [by the committee sans HOD-CSIS]

ii. Final approval- [by HOD-CSIS]

iii. Publish the DDF balance sheet once a year to the department faculty members. [by HOD-CSIS]

# III. Research Support [Max 30% of C-DDF and 30% W-DDF]

Department DDF will be used to support conference registration, travel, journal over-length page charges (EPC), **subject to the availability of funds**. The DDF scrutiny committee will perform the selection by following the broad guideline below.

- The support is primarily for accepted papers and the support will be in the form of a “Top-up” subject to conditions mentioned in Table 1. The top-up can be used for registration, travel and accommodation.
- The University also charted a policy for OA Journal Page charges. In view of this, ***DDF will not have any allocation for OA Journals.***
- There will be top-up support for the following three categories:
- **[TOP] Top tier:**
  ✓ **Jour:**This support is provided when there is no university wide support from BITS yet. This support will be withdrawn once there is a university wide support from BITS.
  ***SciMago Q1, IEEE and ACM Transactions*** journals (for overlength page charges only)
  ***Inter-disciplinary society journals*** whose impact factors are > 5 and is in Q1.
  ✓ **Conf:** Conference venues (Main Track Only) in ***{CSRankings UNION CORE-A*}.***
- **[OTH]**This will be decided by the department level DDF scrutiny committee based on the merit of the paper and the venue. There are following four subcategories:
  **OTH-1.** ***CORE A main track*** paper
  **OTH-2.**
  i) ***CORE A* parallel track*** (workshop, symposium, industry tracks **NOT** considered) **AND**
  ii) it is old enough, i.e. running for **at least 5 years AND**
  iii) steering committee comprising of **reputed scholars**. Examples of such tracks are: ICSE-SEIP, ICSE-SEET, ICSE-MSR etc.
  **OTH-3.** ***CORE B main track***: for **Non-AI** subject areas only
  **OTH-4.** ***Pen and Pencil Workshops***: DDF shall support participation in a few selected workshops of this kind provided the faculty is one among the invited scholars, selected based on their contributions in the topic or area. The department encourages such participations, since an outcome of these workshops, many times ends up in having a publication at prestigious venues. The following are the selected venues (will be revised time to time):
  I. Dagstuhl
  II. Lorentz Center
  III. Banff, including Banff at CMI, Chennai.
  IV. American Mathematical Society
  V. Hausdorff Research Institute for Mathematics
  VI. Simons Institute at UC Berkeley

Supporting documents: Faculty should provide a **letter of invitation** and a list of scholars (with affiliations) participating in the program.

- **[OTH-Students]** DDF may extend conference registration support to papers **authored** by all full time students only (no faculty member as co-author) for
  a) ***CORE A* workshop, symposium, industry tracks*** (all areas) as well as ***CORE A workshop, symposium, industry tracks*** for **non-AI subject** areas
  b) The venue should be old enough, i.e. running for at least 5 years
  c) steering committee comprising of **reputed scholars**.
- If a conference paper is selected for funding through DDF, the funding given to a faculty can be used for
  - Registration
  - Travel (typically the registered author, or a co-author from BITS Goa only)

The top-up matrix and the selection guideline for conferences is shown in Table 1 .

*Table 1 Topup Matrix*

| Category | Eligibility | Frequency | Criteria | Max Funding |
|---|---|---|---|---|
| [TOP]. Conf. | 1. All faculty members and Visiting Full-time faculty.<br>2. Full-time Students* - AT LEAST one author is a faculty member from CS&IS department and the faculty member chooses not to avail DDF for the same article. | Always eligible, no limit | Criteria defined in TOP.Conf | 1.5 Lakhs INR; EVEN if PDA is NOT exhausted |
| [TOP]. Jour | | | For journals belonging to TOP.Jour | Max USD 500; ONLY if PDA is exhausted |
| [OTH] | | An author is eligible to apply in this category once a year. | Criteria defined in OTH-1 | 80000 INR; EVEN if PDA is NOT exhausted |

| | | | | |
|---|---|---|---|---|
| | 1. Preferably Assistant Professors and Visiting Full-time faculty.<br>2. Full-time Students - AT LEAST one author is a faculty member from CS&IS department and the faculty member chooses not to avail DDF for the same article. | | Criteria defined in OTH-2 | 60000 INR; ONLY if PDA is exhausted |
| | | | Criteria defined in OTH-3 | 50000 INR; ONLY if PDA is exhausted |
| | | | Criteria defined in OTH-4 | 40000 INR; ONLY if PDA is exhausted |
| [OTH-Students] | Full-time Students - authors are students only, NO faculty member, or NO industry mentor as co-author | An author is eligible to apply in this category once a year. | Criteria defined in OTH-Students | 50000 INR |
| Encouragement | Assistant Professors, Visiting Full-time faculty and Ph.D. Students ONLY | Based on the event. One travel per conference without any paper. | A*/A conferences PLUS a few approved conferences (see Appendix A). | 25000 INR |

*Full-time students refers to all students enrolled in undergraduate, masters and full-time PhD students

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**IV. Conference Debriefing: Whoever (faculty/student) is supported by DDF should give a presentation of their work after coming back from the conference. The individual supported by the DDF is expected to book the CSIS meeting room (or other room if this is unavailable) for one hour (45 min + 15 min for questions) and invite all the PhD students and faculty. The presentation shouldn't just be his/her own work but about the conference and the learnings.**

a) What's the conference about, its focus, why is it great?

b) What's the focus area/theme this year?

c) About the location. What was the submission volume, acceptance rate, geo distribution of papers, who else from India (or BITS)?

d) What are the workshop themes? They give a clue of the new advancements, upcoming research ideas

e) Lessons learnt from the Keynote talks

f) What do you think is the trend/ or what are your biggest takeaways from this conference

g) What did you present - your work in brief, how did they receive your work (max 15 min)?

h) Describe one paper you liked in this conference (10min)

i) Any possible collaboration?

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## Note:

1. For **OTH** category conferences (OTH-2, OTH-3, OTH-4), the policy of DDF support as described in the matrix, applies **only when** the PDA is exhausted for a regular faculty. However, PDA restriction doesn't apply to Visiting Full time faculty.
2. The 'Non-AI subject areas' classification is intended to bring parity between AI and Non-AI domains.
3. **Under Encouragement Category**, Ph.D. students are allowed for Indian Conferences only. Under the same category, faculty members (visiting and assistant professors) are eligible for CORE A/A* and a few additional conferences mentioned in Appendix A, BUT NOT for any Indian Conference.
4. If the authors anticipate to use the DDF based on the criteria, they are expected to acknowledge the CSIS department

**Appendix A Approved Conferences (over and above A/A* conferences)**

| Name | Applicable to |
|---|---|
| CLEAR | Faculty |
| ALT | Faculty |
| DeepMath | Faculty |
| ICPRAI, CHIL | Faculty |
| Indian Conferences:<br>IndoCrypt, IndoML, ICDCN, COMSNET, ISEC, CODS-COMADS, AI-ML Systems, SaTML, ICASSP, HRI, FSTTCS, HIPC | Ph.D. students only |

This list is subject to a re-evaluation once in each year by the DDF scrutiny committee.

# FDCM Policy

TAs can be termed as FDCM (First Degree Course Mentor) under the Departmental FDCM Programme (informal TA's).

**The details of the Programme will remain the same as below:**

Details of the Departmental FDCM programme:

1. The FDCM programme will be managed by the Department.
2. Selection of FDCMs will be done by the Instructor-in-Charge, who will complete entries in an electronic document that lists: (a) IC; (b) Course; (c) Student; (d) Expected duties; and (e) Performance Assessment (see 4 below).
3. **Student duties cannot include evaluation.**
4. Performance Assessment will be completed by the IC at the end of the course. It will consist of 2 parts: (a) Recommendation; and (b) Comments. "Recommendation" will be one of the following: Honorarium+Certificate (H+C); Certificate only (C only); None (N). "Comments" will be (short) free text explaining the recommendation.
5. Honoraria will have the following constraints: (a) a honorarium cannot be recommended for a student already receiving a payment under the institute's FDTA programme; (b) The number of FDCMs recommended for an honorarium cannot exceed a fixed ratio to the number of students in the course (currently 1:20); (c) The honorarium for an FDCM cannot exceed a fixed amount (currently NR 2,000); (c) Total payments will be subject to a Departmental cap per semester (currently INR 50,000); (d) if the recommendations for payments in a semester exceed the Departmental cap, then the number of students eligible for honoraria will be reduced accordingly; and (e) The final honorarium for all FDCMs will be the same, and will be transferred to the student through the SWD.
6. The Certificate will not distinguish between different categories of performance. It will contain something like the following: (a) Title: "Certificate of Appreciation"; (b) followed by "Awarded to: X"; (c) "for excelling as a Course Mentor in the subject: Y"; (d) Department details, date, signatures etc.
7. The recommendation of categories (H+C; C only; N), will be done by the IC based on assessment of performance, and other criteria (whether eligible for an honorarium) etc.
8. The FDCM program will use the Departmental Development Fund for honoraria and sundry expenses (like printing certificates).
9.

# FD/HD Thesis Policy

## CSIS OFF CAMPUS THESIS

### Background

The department strongly recommends that students consider collaborating with CSIS faculty members at the institute for their bachelor's or master's thesis. While students have the flexibility to select another institution or corporation for their thesis, the department adheres to the institute's regulations for off-campus thesis.

It is assumed that students opt for an off-campus thesis rather than working with on-campus faculties because they believe the off-campus work better aligns with their interests and of course, the work content meets the criteria for a "thesis".

## Guidelines

Based on the above background, specific guidelines are established to uphold the essence of an off-campus thesis laid by BITS Pilani. For off-campus thesis, a CSIS student must select an appropriate CSIS faculty member as their on-campus mentor. Furthermore:

1. The topic must be mutually interesting to both on-campus and off-campus supervisors.
2. Regular progress updates must be provided to the off-campus supervisor through presentations. Failure to do so will be interpreted as a lack of progress from the student.
3. The thesis must address a set of research or engineering challenges that necessitate thorough investigation and subsequent resolution by the student. A robust implementation and evaluation of the solution are also anticipated. Refer to the Appendix for the TOC of a thesis.
4. The thesis content should be submission worthy to a reputable conference or journal (Scopus indexed), with the quality of the venue determined jointly by the off-campus and on-campus supervisors.

CSIS Faculty members acting as on-campus supervisors, should ensure that work proposal adheres to the guidelines mentioned above. Thesis evaluation by the on-campus supervisor should be based on points 2, 3, and 4.

## Application Process:

1. Students must obtain initial approval from an on-campus CSIS faculty member.
2. Subsequently, the faculty member should email the Head of Department (HOD) detailing the proposal's merits and attaching the signed approval form. Direct emails from students will not be entertained.

3. Dual Degree Students: For a dual degree student, the on-campus faculty should be from CSIS department and the other primary department with which the student is enrolled.

## Exceptional Circumstances:

Choosing a faculty member from another department as the main on-campus mentor of the thesis (which is obviously a core CSIS topic) is an exception. It requires approval from the CSIS HOD, as per the institute policy.

**Such a case is strongly discouraged and generally NOT approved.**

## Appendix: What makes a good thesis (on campus or off campus)?

A good thesis typically demonstrates a student's ability to conduct independent study, critically analyze information, and communicate their findings effectively. Some key components of a good thesis include:

1. **Clear research (or engineering) question or objective:** The thesis should clearly state the research question or objective it aims to address.
2. **Literature review:** A thorough review of existing literature related to the research topic is essential to demonstrate an understanding of the existing knowledge in the field.
3. **Methodology:** A well-defined and appropriate evaluation methodology should be used to collect and analyze data, or existing results. Subsequenty, it should describe how the solution has been i
4. **Analysis and take-aways:** The thesis should present a clear interpretation of the results in relation to the research question.
5. **Structure and organization:** A well-organized and structured thesis with clear sections for introduction, literature review, methodology, results, discussion, and conclusion is important for readability and coherence.
6. **Critical thinking:** The thesis should showcase the student's ability to think critically, evaluate evidence, and draw meaningful conclusions.
7. **Contribution to the field:** A good thesis should aim to make a meaningful contribution to

the field of study, whether through new insights, practical implementation (with clearly demonstrated benefits), or theoretical advancements.

8.

# PhD Hall Policy

The keys to the following PhD Hall Rooms - D-152, D-154, D-251. will now be **stored in the ESC Room (D-252)** and **must not be returned to the Security Key Room** under any circumstance.

To ensure smooth access and accountability, a **Key Register** has been placed in the ESC Room (D-252) to record **IN** and **OUT** entries of the keys.

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### All PhD Scholars are required to strictly follow the guidelines below:

1. **Mandatory Register Entry:** Anyone collecting or returning a key must compulsorily log the date, time, name, and signature in the key register.
2. **No Off-Campus Key Carrying:** Keys must remain on campus at all times and are **not to be taken to hostels or homes.**
3. **Responsibility for Return:** The last person using the room is responsible for returning the key to the ESC Room (D-252). In case of loss or misplacement, the person who last signed out the key will be held accountable.
4. **Access Restrictions:** Only the PhD Scholars officially allotted a particular room are permitted to use it. No one else should enter or use that room without prior approval.
5. **Security and Cleanliness:** Please ensure rooms are locked when unoccupied and maintained in a clean, orderly state.
6. **For Access Requests:** Scholars who currently do not have access to **Esc Room** may contact **Shreenivas, Neha, or Aditi** for assistance.

**For any clarifications, feel free to reach out to the CSIS Office.**

# Streamlining the Project Course Workflow

Project courses provide an excellent platform for collaboration between faculty members and students. However, the process of applying for and approving these courses has been quite ad-hoc. Many students lacked clarity regarding the course policies and were often unaware of which faculty members were offering what projects.

To address these issues and bring more structure to the process, we have taken the following steps:

- The project-type course policy has been clearly outlined and is now available on the CSIS intranet: [View Project Course Policy]
- To further enhance transparency, we have also published the past project proposal data (from AUGSD) on the intranet. This gives students insight into previously approved projects and the kinds of work faculty members typically supervise: [View Past Project Proposals (SoP/DoP/LoP/SaT)]

# New Workflow for Project Course Proposals

We have introduced a new application that simplifies and structures the process of offering and applying for project courses. Here's how it works:

- Faculty members can log in to the application and propose one or more project courses.
- Students can then log in, browse all the projects floated by faculty members, and apply to the ones they are interested in.
- Faculty members review the applications they receive and approve or reject them accordingly.
- The application uses ACM CCS taxonomy to tag a particular project proposal, making it easier for students to discover projects aligned with their interests.
- The application allows the admin to download a CSV file containing details of all approved projects.
- Our current ERP system does not support a seamless integration with this application. Therefore, the students will have to manually apply to ERP and the faculty member will still need to manually approve these projects in the ERP.

# Course Taken

Faculty Name:

All A BaskarAbhisek ChakrabartyAditya ChallaAmitava DasAnand Narashimha MurtyAnand NarasimhamurthyAniket Basu RoyAnup Basil MathewArnab Kumar PaulAshwin SrinivasanBasabdatta BhattacharyaBharat M. DeshpandeBiju K. RaveendranDevashish GosainDiptendu ChatterjeeGargi Sanket PrabhuHarikrishnan N. B.Hemant RathoreKanchan MannaKunal Kishore KorgaonkarManideepa Mukherjee BhakatNeena GoveasRajesh KumarRamprasad S. JoshiRaunakRizwan ParveenSanjay Kumar SahaySantonu SarkarShubhangi K. Gawali.Siddharth GuptaSnehanshu SahaSougata SenSravan DandaSujith ThomasSurjya GhoshSwaroop Ravindra JoshiSwati AgarwalTanmay Tulsidas VerlekarVinayak Naik

Semester:

All Sem I 2022-2023Sem I 2023-2024Sem I 2024-2025Sem I 2025-2026Sem II 2022-2023Sem II 2023-2024Sem II 2024-2025semid

| Faculty | Course Name | Sem |
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| | LOGIC IN COMPUTER SC | Sem I 2023-2024 |

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| | ADVANCED ALGORITHMS AND COMPLEXITY | Sem I 2024-2025 |
| | LOGIC IN COMPUTER SC | Sem I 2025-20 |

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| | DATA STRUCTURES & ALGO | SemII2022-2023 |

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| Abhisek Chakrabarty | NATURAL LANGUAGE PROCESSING | Sem I 2025-2026 |
| Aditya Chal | APPLIED MACHINE LEARNING | Sem I 2022-2 |

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| | MACHINE LEARNING | Sem I 2024-20 |

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| | COMPUTATIONAL LEARNING THEORY | Sem I 2025-2026 |
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| | COMPUTER PROGRAMMING | Sem II 2023-20 |

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| Amitava Das | Information Retrieval | Sem I 2025-2026 |
| Anand Na | MACHINE LEARNING | Sem II 20 |

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| AnandNarasimhamurt | MACHINE LEARNING | SemI2025-2026 |

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| Aniket Basu Roy | Approximation Algorithms | Sem II 2024-2025 |

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| Anup Basil Mathew | THEORY OF COMPUTATION | Sem I 2022-2023 |
| | COMPUTER PROGRAMMING | Sem II 2022-20 |

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| Arnab Kumar Paul | OPERATING SYSTEMS | Sem I 2022-2023 |

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| | DATA STORAGE TECHNOLOGIES AND NETWORKS | Sem I 2022-2023 |
| | COMPUTER ARCHITECTURE | Sem I 2023-20 |

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| | DATA STORAGE TECHNOLOGIES AND NETWORKS | Sem I 2023-2024 |

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| | DATA STORAGE TECHNOLOGIES AND NETWORKS | Sem I 2024-2025 |
| | OPERATING SYSTEMS | Sem I 2025-20 |

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| | COMPUTER PROGRAMMING | Sem II 2022-2023 |

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| Ashwin Srinivasan | LOGIC IN COMPUTER SC | Sem I 2022-2023 |

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| | MACHINE LEARNING | Sem I 2023-20 |

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| | MACHINE LEARNING | SemII2 |

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| Basabdatta Bhattachar | OBJECT ORIENTED PROG | Sem I 2022-2023 |

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| | DATABASE SYSTEMS | Semll2022-20 |

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| | COMPUTATIONAL NEUROSCIENCE | Sem II 2022-2023 |
| | DATABASE SYSTEMS | Sem II 2 |

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| | COMPUTATIONAL NEUROSCIENCE | SemII 2023-2024 |

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| | COMPUTATIONAL NEUROSCIENCE | Sem II 2024-2025 |
| Bharat M. Deshpa | ADVANCED ALGORITHMS AND COMPLEXITY | Sem I 2022-20 |

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| | OBJECT ORIENTED PROG | Sem I 2023-2024 |
| | Discrete Mathematical Structures | Sem I 2025-20 |

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| | THEORY OF COMPUTATION | Sem I 2025-2026 |
| | DESIGN & ANAL OF ALGO | Sem II 20 |

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| | PH D SEMINAR | Sem II 2022-20 |

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| | PH D THESIS | Sem II 2022-2023 |

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| | DESIGN & ANAL OF ALGO | Sem II 2023-20 |

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| | DESIGN & ANAL OF ALGO | SemII2024-2025 |

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| Biju K. Raveendr | OPERATING SYSTEMS | Sem I 2022-20 |

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| | OPERATING SYSTEMS | Sem II 2022-20 |

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| | REAL TIME SYSTEMS | Sem II 2 |

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| | REAL TIME SYSTEMS | Seml 2024-2025 |

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| | OPERATING SYSTEMS | Sem II 2024-2025 |
| Devashish Gosa | SELECTED TOPICS FROM COMPUTER SCIENCE | Sem I 2023-20 |

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| | INFORMATION<br>SECURITY PROJECT | S<br>e<br>m<br>I<br>2<br>0<br>2<br>3<br>-<br>2<br>0<br>2<br>4 |
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| | COMBINATORIAL OPTIMIZATION | Sem II 2024-20 |

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| Gargi Sanket Prabhu | PARALLEL COMPUTING | Sem I 2023-2024 |

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| | COMPUTER ORGANIZATION AND ARCHITECTURE | Sem I 2023-2024 |
| | PARALLEL COMPUTING | Sem I 2024-20 |

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| | PARALLEL COMPUTING | Sem I 2025-2026 |
| | MICROPROC & INTERFACING | Sem II 20 |

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| | MICROPROC & INTERFACING | SemII2024-2025 |

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| Harikrishnan N.B. | BIO-INSP INTELLI: ALGO & APPL | Sem I 2022-2023 |
| | OBJECT ORIENTED PROG | Sem I 2022-20 |

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| | Introduction to Data Science | Sem II 2022-20 |

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| | BRAIN-INSPIRED DEEP LEARNING | SemII2023-2024 |
| Hemant | MACHINE LEARNING | SemI20 |

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| Rathore | | 23-2024 |
| | DATA MINING | SemI2023-2024 |

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| | ADVANCED DATA MINING | Sem I 2025-20 |

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| | COMPUTER PROGRAMMING | Sem II 2022-2023 |
| | OPERATING SYSTEMS | Sem II 2 |

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| | INTRODUCTION TO DATA SCIENCE | Sem II 2023-2024 |

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| | DATA MINING | Sem II 2024-2025 |
| Kanchan Man | COMPUTER ARCHITECTURE | Sem I 2022-20 |

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| | COMPUTER ARCHITECTURE | Seml2023-2024 |
| | COMPUTER ARCHITECTURE | Seml202 |

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| | COMPUTER ARCHITECTURE | Seml2025-2026 |

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| | ADV COMPUTER ARCHITECTUR | Sem II 2022-20 |

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| | ADV COMPUTER ARCHITECTUR | SemII2023-2024 |

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| | ADV COMPUTER ARCHITECTUR | Sem II 2024-2025 |
| Kunal Kishore Ko | PRINCIPLES OF PROGG LANG | Sem I 2022-20 |

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| | QUANTUM ARCHITECTURE AND PROGRAMMING | Sem I 2024-20 |

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| | QUANTUM ARCHITECTURE AND PROGRAMMING | Sem I 2025-2026 |
| | DISTRIBUTED COMPUTING | Sem I 202 |

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| | QUANTUM ARCHITECTURE AND PROGRAMMING | Sem II 2022-2023 |

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| | COMPILER CONSTRUCTION | Sem II 2022-2023 |
| | ARTIFICIAL INTEL FOR ROBOTICS | Sem II 2023-20 |

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| | QUANTUM ARCHITECTURE AND PROGRAMMING | Sem II 2023-2024 |
| | SYSTEM SECURITY | Sem II 2 |

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| Manideep Mukherjee Bhak | HUMAN COMP INTERACTION | Sem I 2025-2026 |

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| | OBJECT ORIENTED PROG | Seml2025-2026 |

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| | SOFTWARE FOR EMBEDED SYS | Sem II 2022-2023 |
| | SOFT FOR EMBED SYS | Sem II 2022-20 |

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| | DATA SCIENCE FOR HEALTHCARE | SemII2024-2025 |
| Rajesh | LOGIC IN COMPUTER SC | SemI20 |

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| Kumar | | 24-2025 |
| Ramprasad S. Joshi | DISCR STRUC FOR COMP SCI | Seml2022-2023 |

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| | OBJECT ORIENTED PROG | Sem I 2023-2024 |
| | COMPUTATIONAL GEOMETRY | Sem I 2023-20 |

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| | NATURAL LANGUAGE PROCESSING | Sem I 2023-2024 |
| | PRINCIPLES OF PROGG LANG | Sem I 202 |

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| | PRINCIPLES OF PROGG LANG | Seml2025-2026 |

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| | COMPUTATIONAL GEOMETRY | Sem II 2022-2023 |
| | COMPILER CONSTRUCTION | Sem II 2022-20 |

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| | DEV & USE OF COMP SOFT | SemII2023-2024 |

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| | COMPILER CONSTRUCTION | Sem II 2023-2024 |
| | COMPILER CONSTRUCTION | Sem II 2024-20 |

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| Raunak | DEEP LEARNING | Seml2025-2026 |
| Rizwan P | PRINCIPLES OF PROGG LANG | Seml202 |

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| | COMPUTER PROGRAMMING | Sem II 2022-2023 |
| | SOFTWARE ARCHITECHTURES | Sem II 2022-20 |

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| Sanjay Kumar Sahay | CRYPTOGRAPHY | Sem I 2022-2023 |

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| | INFORMATION SECURITY PROJECT | Sem I 2022-2023 |
| | BLOCKCHAIN TECHNOLOGY | Sem I 2023-20 |

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| | OPERATING SYSTEMS | Sem I 2023-2024 |
| | CRYPTOGRAPHY | Sem I 202 |

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| | BLOCKCHAIN TECHNOLOGY | Seml2025-2026 |

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| | CRYPTOGRAPHY | Sem I 2025-2026 |
| | COMPUTER NETWORKS | Sem II 2022-20 |

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| | NETWORK SECURITY | SemII2022-2023 |
| | NETWORK SECURITY | SemII2 |

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| | COMPUTER NETWORKS | Sem II 2024-2025 |

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| Santonu Sarkar | CLOUD COMPUTING | Sem I 2023-2024 |
| | DISTRIBUTED COMPUTING | Sem I 2024-20 |

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| | Programming Languages and Compiler Construction | Sem I 2025-2026 |
| | COMPILER CONSTRUCTION | Sem II 20 |

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| | CLOUD COMPUTING | SemII2023-2024 |

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| | CLOUD COMPUTING | Sem II 2024-2025 |
| Shubhangi K. Gaw | OPERATING SYSTEMS | Sem I 2022-20 |

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| | ADVANCED OPERATING SYSTEMS | S<br>e<br>m<br>I<br>2<br>0<br>2<br>2<br>-<br>2<br>0<br>2<br>3 |

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| | ADVANCED OPERATING SYSTEMS | Sem I 2023-20 |

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| | OPERATING SYSTEMS | Sem I 2024-2025 |
| | Advanced Database Systems | Sem I 202 |

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| | DATABASE SYSTEMS | Seml2022-2023 |

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| Siddharth Gupta | DISCR STRUC FOR COMP SCI | Sem I 2024-2025 |

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| | DISCR STRUC FOR COMP SCI | Sem I 2025-2026 |
| | COMPUTATIONAL GEOMETRY | Sem II 2023-20 |

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| | COMPUTATIONAL GEOMETRY | SemII2024-2025 |

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| Snehanshu Saha | COMPUTATIONAL LEARNING THEORY | Sem I 2022-2023 |
| | DISCR STRUC FOR COMP SCI | Sem I 2023-20 |

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| | FOUNDATIONS OF DATA SCIENCE | Seml2024-2025 |
| | FOUNDATIONS OF DATA SCIENCE | Seml202 |

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| | FOUNDATIONS OF DATA SCIENCE | Seml2022-2023 |

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| | FOUNDATIONS OF DATA SCIENCE | Sem II 2023-2024 |
| | Project on Social and Environmental Applications of Data Science | Sem II 2023-20 |

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| | FOUNDATIONS OF DATA SCIENCE | SemII2024-2025 |

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| | Project on Social and Environmental Applications of Data Science | Sem II 2024-2025 |
| Sougata S | COMPUTER ARCHITECTURE | Sem I 2022-20 |

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| | INTERNET OF THINGS: DESIGN AND DEVELOPMENT | Sem I 2025-20 |

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| | COMPUTER NETWORKS | SemII2022-2023 |
| | COMPUTER NETWORKS | SemII2 |

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| | | 023-2024 |
| | NETWORK SECURITY | SemII2024-2025 |

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| Sravan Danda | APPLIED STATISTICAL METHODS | Sem I 2022-2023 |
| | ADVANCED STATISTICAL TECHNIQUES FOR ANALYTICS | Sem I 2023-20 |

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| | REAL TIME SYSTEMS | Sem I 2024-2025 |
| | APPLIED STATISTICAL METHODS | Sem I 202 |

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| | | 4-2025 |
| | NATURAL LANGUAGE PROCESSING | Seml2024-2025 |

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| | APPLIED STATISTICAL METHODS | Sem I 2025-2026 |
| | ARTIFICIAL INTELLIGENCE | Sem II 2022-20 |

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| | REINFORCEMENT LEARNING | SemII2023-2024 |

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| | APPLIED STATISTICAL METHODS | Sem II 2024-2025 |
| Sujith Thom | ARTIFICIAL INTELLIGENCE | Sem I 2022-20 |

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| | ARTIFICIAL INTELLIGENCE | Sem I 2025-2026 |

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| | DATA STRUCTURES & ALGO | Sem II 2022-2023 |
| | READING COURSE | Sem II 2023-20 |

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| | DATA STRUCTURES & ALGO | Sem II 2023-2024 |

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| | DATA STRUCTURES & ALGO | Sem II 2024-2025 |
| Surjya Gho | PERVASIVE COMPUTING | Sem I 2022-20 |

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| | HUMAN COMP INTERACTION | Sem II 2022-20 |

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| | OPERATING SYSTEMS | SemII2022-2023 |

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| | COMPUTER PROGRAMMING | Sem II 2023-2024 |
| | HUMAN COMP INTERACTION | Sem II 2024-20 |

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| | | 25 |
| Swaroop Ravindra Joshi | OBJECT ORIENTED PROG | Sem I 2022-2023 |

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| | SOFTWARE DEVELOPMENT FOR PORTABLE DEVICES | Sem I 2022-2023 |
| | OBJECT ORIENTED PROG | Sem I 2023-20 |

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| | SOFTWARE ENGINEERING | Sem I 2023-2024 |

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| | SOFTWARE DEVELOPMENT FOR PORTABLE DEVICES | Sem I 2023-2024 |
| | SOFTWARE DEVELOPMENT FOR PORTABLE DEVICES | Sem I 2024-20 |

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| | COMPUTER PROGRAMMING | Sem I 2025-2026 |
| | COMPUTER PROGRAMMING | Sem I 202 |

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| | SOFTWARE DEVELOPMENT FOR PORTABLE DEVICES | Seml2025-2026 |

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| | COMPUTER PROGRAMMING | Sem II 2022-2023 |
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| Swati Agarwal | OBJECT ORIENTED PROG | Seml2022-2023 |
| | NATURAL LANGUAGE PROCESSING | Seml202 |

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| | SOCIAL MEDIA ANALYTICS | Seml2023-2024 |

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| | COMPUTER PROGRAMMING | Sem II 2022-20 |

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| Tanmay Tulsidas Verlekar | APPLIED MACHINE LEARNING | Sem I 2024-2025 |

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| | GENERATIVE AI | Sem I 2024-2025 |
| | GENERATIVE AI | Sem I 2025-20 |

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| | SELECTED TOPICS FROM COMPUTER SCIENCE | Sem II 2022-2023 |
| | DEEP LEARNING | Sem II 2 |

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| | DEEP LEARNING | SemII2023-2024 |

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| | DEEP LEARNING | Sem II 2024-2025 |
| Vinayak Na | ADVANCED COMPUTER NETWORKS | Sem I 2022-20 |

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| | SELECTED TOPICS FROM COMPUTER SCIENCE | Sem I 2023-2024 |
| | ADVANCED COMPUTER NETWORKS | Sem I 2024-20 |

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| | ADVANCED COMPUTER NETWORKS | Sem I 2025-2026 |
| | NETWORK PROGRAMMING | Sem II 20 |

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# Policy for selecting Project type courses: SOP/DOP/LOP

## Preamble: Existing BITS Policy

In addition to discipline electives, BITS offer a set of project-type courses. Accordingly, the CSIS department at the Goa Campus offers the following project-type courses, which may be (not compulsorily) taken by the students to meet the discipline elective requirements.

- CS F266 Study Project
- CS F366 Laboratory Project
- CS F367 Laboratory Projects
- CS F376 Design Project
- CS F377 Design Project
- CS F491 Special Project

For example, the CS F266 Study Project is intended for a student of B.E. CSIS and is considered a CSIS department elective.

<u>To ensure a project course meets the Discipline Elective Requirement under the head of (Discipline) Electives, the following conditions apply:</u>

(a) All of these Project courses should be

(i) within the Discipline (for which the degree is being awarded) or
(ii) from an allied Discipline if so specified by the Department offering the degree

(b) The projects may be chosen from under these sub-heads.

- Study Projects (maximum of 1)
- Laboratory (maximum of 2)
- Design Projects (maximum of 2)
- Special Projects (maximum of 1)

An FD student may avail of a maximum of 3 Project courses (under any of the heads mentioned above offered by any discipline as an Open Elective. However, in total, an FD student may avail at most 5 Project courses against elective slots in any category.

# CSIS Policy on CSFXXX Project Courses

In view of the above rule defined by BITS, CSIS enforces the following:

1. If a CSIS student takes any of the CS FXXX project type courses, it HAS TO be within CSIS (a.i) or from the allied disciplines (a.ii), which is EEE or Maths. Rationale: Note that EEE teaches quite a few CS courses (with CS course code), and so does CSIS; hence, EEE is naturally allied with CSIS. A similar logic applies to the Mathematics department as well.
2. A CSIS student is free to choose project courses offered by other disciplines as an OPEN elective
3. If a CSIS student decides to take a CS FXXX project course as a department elective, this course/project MUST BE supervised by a CSIS faculty (regular, visiting, adjunct or

Professor of Practice), because it is after all a department elective. The student can always take a co-supervisor from another discipline.

4. If a CSIS student takes a CS FXXX project course where the supervisor is from another discipline, it deviates from the norm, and it is treated as an EXCEPTION. **Such an exception requires HoD approval. This is discouraged and not approved almost always.**

# New Workflow for Project Course Proposals

We have introduced a new application that simplifies and structures the process of offering and applying for project courses. Here's how it works:

- Faculty members can log in to the application and propose one or more project courses.
- Students can then log in, browse all the projects floated by faculty members, and apply to the ones they are interested in.
- Faculty members review the applications they receive and approve or reject them accordingly.
- The application uses ACM CCS taxonomy to tag a particular project proposal, making it easier for students to discover projects aligned with their interests.
- The application allows the admin to download a CSV file containing details of all approved projects.
- Our current ERP system does not support a seamless integration with this application. Therefore, the students will have to manually apply to ERP and the faculty member will still need to manually approve these projects in the ERP.

To further enhance transparency, we have also published the past project proposal data (from AUGSD) on the intranet. This gives students insight into previously approved projects and the kinds of work faculty members typically supervise: [View Past Project Proposals (SoP/DoP/LoP/SaT)]

# Instructions on how to use this application

## Assignment of TAs for our Courses

The TA allocation to a course is based on the principle of stable marriage and is a modified form of Gale & Shapley's algorithm.

**What forms a stable marriage in our case?**

The algorithm takes in a list of students and courses with their preferences for the courses and students, respectively. It gives a stable allocation based on a student's preference for a course and a course's "preference" for a student. In our stable allocation, there is no such scenario:

- **Student s1 gets a TAship for Course c1 but actually prefers c2**
- **A Course c2 gets a Student s2 as a TA, but it actually "prefers" another Student s1**

We selected stable marriage allocation since it has one of the highest welfare gains of all allocation algorithms (it means it's good).

**Our solution works in two phases.**

1. **Generate TA Quotas for each course**
2. **Allocate TAs to a course.**

---

**1. Precondition**

The application is executed only after course allocation to all the faculty members is complete.

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**2. Generate TA Quotas.**

Generation of TA quota means that the algorithm decides how many TAs (FD, HD, PhD together) are required for this course. For this purpose, it uses the following attributes of a course:

a) Course Type: CDC without lab, CDC with Lab, Elective, Elective with Lab/Assignment, First Degree or Higher Degree or WILP

b) Typical course strength *(n)*: Since we have to allocate TAs even before registration, it is impossible to know the exact course strength. In view of this, we use historical enrollment data starting from the previous semester.

Each course (c) attribute has a weight (w) associated with it, which plays a role to decide the number of TAs required. A CDC with Lab has the highest weightage, followed by Elective with Lab => CDC without Lab => Elective.

First, the normalized weight of course c is calculated: wavg_c = (n_c*w_c)/sum (n_i*w_i) for all courses. Next, the TA quota is calculated as Total_TA*wavg_c.

For instance, there are a total of 4 courses. Course A is CDC with lab and w_A= 1.5. Suppose that this course has 120 students enrolled. Thus, n_A*w_A = 120*1.5=180. Suppose that n*w factor of the other three courses, B, C, and D, are 180, 150, and 120, respectively. In that case, the normalized weight for A wavg_A = 180/(180+180+150+120)=0.286. Suppose that we have a total 20 FDTA available for the department. The course A gets 20*0.286=5.72 i.e. 6 FDTAs. We apply a similar computation to obtain the number of HD and PHDTAs for course A, once we know the total number of HD and PHDTAs available.

Other factors exist, such as the fact that FDTAs can't be assigned to an HD course. A first degree has more weightage for FDTAs than HDTAs and PhDTAs. These are also considered while computing the quota. We do not elaborate them in detail for brevity.

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**3. Allocate TAs to a course**

**Students' preferences** are taken using a web-based GUI (check our intranet). A student has 5 preferences in decreasing order, spread across FD and HD level courses.

**Course preferences** are computed based on the **CGPA** and **subject grade** of that student in that particular course. If two or more FD students have identical CGPA and subject grades, the tie is broken randomly.

**3.1 Steps**

1. Round 1:

    a. [propose] The algorithm tries to allocate a course to each student based on his/her first preference, and each subject gets some proposals.

    b. [choose] Each subject does its choosing based on its preference, with each subject choosing the top k number of students as it's provisional TAs; where k is the quota of TAs allocated to it.

    c. [reject] The students rejected by subjects go to the next round.

2. Round 2:

    a. [propose] the students propose their second preferences in the next round.

    b. [choose] Now, the subjects again do choosing, mixing the new students with its provisional TAs, and again selecting top k students.

    c. [reject] The students rejected here go to the next round.

3. The algorithm terminates when it runs out of students to proceed to the next round or when the students in the proposing pool have exhausted their filled preferences (that is, they are rejected by all the filled subjects).

The abovementioned steps are applied to FDTAs, HDTAs, and PHDTAs individually. Since no CGPA or subject grade data is available for an HD or a PhD student, a course has NO preference for an HD or a PhD student. Thus, for PhD and HD students, the algorithm first allocates a course to an HD/PHD student based on his/her first preference. Next, a course will randomly select a set of HD/PHD students from its list of proposals. The rejected students go to the next round for their second preference and so on.

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**4. Final Allocation**

There is no guarantee that every student will get a TAship or a course will get a TA. Thus, it is possible that a course does not get any TA at all or that a student remains unallocated. The solution's output is reviewed, and such corner cases are handled manually before submitting the allocation to the instructions office.

# Ph.D. Thesis Submission Process through ERP after completion of Draft thesis reviewed by the DAC member.

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## For Students

1. The Name & ID No in all the documents should be as per the ERP Record.

2. Thesis & Synopsis Should be as per the given below Image.

# Identification of Colorectal Cancer Associated Key Pathways and Potential Biomarkers by Integrated Bioinformatics Analysis

**THESIS**

Submitted in partial fulfillment
of the requirements for the degree of
**DOCTOR OF PHILOSOPHY**

by

**Student Name**
ID No. XXXXXXXXXXXX

Under the supervision of

**Name**

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE,**
**PILANI**
**2024**

**Identification of Colorectal Cancer Associated Key Pathways and Potential Biomarkers by Integrated Bioinformatics Analysis**

**Synopsis**

Submitted in partial fulfillment
of the requirements for the degree of
**DOCTOR OF PHILOSOPHY**

by

**Student Name**
ID No. XXXXXXXXXXXX

Under the supervision of

**Name**

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI**
**2024**

3. List of Publication & references should be added in synopsis.

4. Upload all the documents in ERP as per the checklist given below

1. Thesis
2. Synopsis
3. Two DAC member reports
4. Pre-submission presentation notice collected by DRC
5. Thesis submission Fee Receipt
6. Supervisor Concern Form

7. Turnitin Report

8. Thesis Form I & Thesis Form II

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## For Supervisor.

1. Please verify all the documents
2. Get Approval of Examiner List from DRC.
3. Once Examiner List is approved by the DRC upload all the Examiner details in ERP.
4. Signed documents of Examiner list by DRC & send it to DRC Convener.

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## For DRC Convenor.

1. Please verify all the documents
2. Upload the Signed Documents of Examiner List in ERP & Approve.

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## For DRC Convenor.

- 1. Please verify all the documents
- 2. Upload the Signed Documents of Examiner List in ERP & Approve.

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## For HoD.

- 1. Please verify all the documents
- 2. Approve.

# Policy for selecting Project type courses: SOP/DOP/LOP

## Preamble: Existing BITS Policy

In addition to discipline electives, BITS offer a set of project-type courses. Accordingly, the CSIS department at the Goa Campus offers the following project-type courses, which may be (not compulsorily) taken by the students to meet the discipline elective requirements.

- CS F266 Study Project
- CS F366 Laboratory Project
- CS F367 Laboratory Projects
- CS F376 Design Project
- CS F377 Design Project
- CS F491 Special Project

For example, the CS F266 Study Project is intended for a student of B.E. CSIS and is considered a CSIS department elective.

To ensure a project course meets the Discipline Elective Requirement under the head of (Discipline) Electives, the following conditions apply:

(a) All of these Project courses should be

(i) within the Discipline (for which the degree is being awarded) or

(ii) from an allied Discipline if so specified by the Department offering the degree

(b) The projects may be chosen from under these sub-heads.

- Study Projects (maximum of 1)
- Laboratory (maximum of 2)
- Design Projects (maximum of 2)
- Special Projects (maximum of 1)

An FD student may avail of a maximum of 3 Project courses (under any of the heads mentioned above offered by any discipline as an Open Elective. However, in total, an FD student may avail at most 5 Project courses against elective slots in any category.

# CSIS Policy on CSFXXX Project Courses

In view of the above rule defined by BITS, CSIS enforces the following:

1. If a CSIS student takes any of the CS FXXX project type courses, it HAS TO be within CSIS (a.i) or from the allied disciplines (a.ii), which is EEE or Maths. Rationale: Note that EEE teaches quite a few CS courses (with CS course code), and so does CSIS; hence, EEE is naturally allied with CSIS. A similar logic applies to the Mathematics department as well.
2. A CSIS student is free to choose project courses offered by other disciplines as an OPEN elective
3. If a CSIS student decides to take a CS FXXX project course as a department elective, this course/project MUST BE supervised by a CSIS faculty (regular, visiting, adjunct or Professor of Practice), because it is after all a department elective. The student can always take a co-supervisor from another discipline.
4. If a CSIS student takes a CS FXXX project course where the supervisor is from another discipline, it deviates from the norm, and it is treated as an EXCEPTION. **Such an exception requires HoD approval. This is discouraged and not approved almost always.**

# New Workflow for Project Course Proposals

We have introduced a new application that simplifies and structures the process of offering and applying for project courses. Here's how it works:

- Faculty members can log in to the application and propose one or more project courses.
- Students can then log in, browse all the projects floated by faculty members, and apply to the ones they are interested in.
- Faculty members review the applications they receive and approve or reject them accordingly.
- The application uses ACM CCS taxonomy to tag a particular project proposal, making it easier for students to discover projects aligned with their interests.
- The application allows the admin to download a CSV file containing details of all approved projects.
- Our current ERP system does not support a seamless integration with this application. Therefore, the students will have to manually apply to ERP and the faculty member will still need to manually approve these projects in the ERP.

To further enhance transparency, we have also published the past project proposal data (from AUGSD) on the intranet. This gives students insight into previously

approved projects and the kinds of work faculty members typically supervise: [View Past Project Proposals (SoP/DoP/LoP/SaT)]

IMP LINKS:(Access using only BITS WIFI)

Approval Note for Conferences and Workshops:

https://intranet.csisbitsgoa.com/document/requisition-forms/Approval_Note_for_Conferences&Workshops_2022.pdf

School Fees Reimbursement:

https://intranet.csisbitsgoa.com/document/requisition-forms/Requisition%20form%20for%20grant%20of%20School%20Fees.pdf

Claiming Merit Scholarship for Wards ofEmployee:

https://intranet.csisbitsgoa.com/document/requisition-forms/Claiming-Merit-Scholarship-for-Wards-of-Employee.pdf

Format for claming reimbusement of PROFESSIONAL ALLOWANCE:

https://intranet.csisbitsgoa.com/document/requisition-forms/Format-for-claming-reimbusement-of-PROFESSIONAL-ALLOWANCE23.pdf

Medical Reimbursement - OPD (Form I):

https://intranet.csisbitsgoa.com/document/requisition-forms/Medical-Reimbursement-OPD_Form-I.pdf

**Committee Members:**

1. Dr. Arnab Paul (Chair)

2. Prof. Vinayak Naik (Co-chair)
3. Dr. Kanchan Manna
4. Dr. Kunal Korgaonkar
5. Prof. Sanjay Sahay
6. Dr. Sougata Sen
7. Shreenivas Naik (Support)
8. Neha Naik (Support)
9. Aditi Borkar (Support)

Members of this committee will perform the responsibilities stated below for at least one year. After one year, we can revisit for modification of responsibilities as well as members on a need basis. The HOD will certainly be involved, but the ownership of the following tasks will be with the committee.

**Duties and Responsibilities:**

1. Envision the infrastructure requirement for teaching and research in the future (not just one year) and create an appropriate budget.
   The committee takes input from the department and creates a requirement.

   **Why:** This will help in having a bigger plan, based on which we can aggressively ask for new labs.
2. 

   Ensure NFA creations for infrastructure procurement and installation on time.

3. Train and demand more manpower for lab support.
4. Monitor the current infrastructure for its appropriate and optimal usage.
    - Bring more automation in monitoring, and create usage data for analysis and prediction.
5. Proactive maintenance:
    - Interacting with the support team
    - Remote management of servers, desktops, software upgrades
6. Asset audit, so that:
    - We know what we have, what needs upgradation
    - Track department-level and individual-level purchases
    - Make decisions to support faculty/students when they require infrastructure
    - Support faculty members in submitting external proposals where infrastructure details are needed

7. Currently, we do not have any central place to get a list of what assets we have, whether we are duplicating the infra, whether the infra can be shared for optimal usage, etc. A faculty member may procure infrastructure for their project and lend it on a need basis.
8. Leverage external/cloud-based resource usage:
    - Actively establish collaboration with IAAS providers like AWS, AZURE, Google (through alumni connect?) for long-term cloud-based infrastructure support
    - Leverage CDAC's supercomputing facility for our research and project use

**Committee Members:**

1. Siddharth Gupta
2. Raunak Manekar
3. Abhisek Chakrabarty
4. Aniket Basu Roy
5. Gargi Sanket Prabhu
6. Hemant Rathore

**Duties and Responsibilities:**

1. PoC for Annual Events like Open Day- coordinate with student clubs, PhD students to showcase various prototypes.

2. Science Day/Engg Day etc. - coordinate with student clubs, PhD students to showcase various prototypes.
3. Research Showcasing event - coordinate with PhD students to showcase various prototypes.
4. Alumni event (to demonstrate our research & academic capability)
5. National and Internal Conferences organized by CSIS (*)
   (*) Conference hosting is typically managed by the local organization chair. However, this team will coordinate with the PhD/ME/FD students to prepare a list of volunteers and assist the local organization chair. They can also participate in managing the event in coordination with the local organization chair if their bandwidth permits.
6. Student Connect: The team should coordinate with the ASCII body to organize an informal gathering with the final year students (BE, ME) for information sharing. This event is separate from the faculty-student interaction organized by the ASCII. The final year students can share the good and not-so-good experiences and suggest ideas for improvements. Likewise, the faculty members can also share their good and not-so-good experiences with the students and suggest ideas for improvements. The feedback can be documented.
   Subsequently,

- DEAC+HoD can then review and propose an implementation plan for the ideas suggested by the students.
- ASCII can review the issues raised by the faculty members, come up with a solution plan and pass them to the junior students.

This event may be organized once a semester, and the event may be funded from the DDF/Dept Budget.

**The team comprises the following people:**

1. Prof. Santonu Sarkar (APAC Chairperson)
2. Prof. Neena Goveas
3. Prof. Sujith Thomas
4. Prof. Swaroop Joshi
5. Prof. Sravan Danda
6. Prof. Arnab Paul
7. ASCII President
8. CSIS Office

**Key Stakeholders:** Faculty, Office staff, and students

**Objective:**

This will be a collaborative initiative between students (ASCII is the primary point of contact) and faculty, focused on enhancing academic workflows through automation. The software aims to automate and streamline various departmental activities.

**There are four primary responsibilities of this committee:**

- **Identify Opportunities:** Identify repetitive or time-consuming tasks, like scheduling, NFA, answer script evaluation/recheck, quizzes, programming assignment evaluation, or form processing.
- **Effort vs. Benefit:** We collect feedback from students and faculty to assess the feasibility and value of a project before undertaking it.
- **Build Projects:** Faculty members define a project or use a Hackathon as a means to develop the software.
- **Version Control with Git:** GIT has been created to manage these projects. The CSIS Office will manage and maintain this software and be responsible for deployment.
- **Maintain and Improve:** The committee should develop a plan for ongoing support, updates, and improvements to ensure the tools remain useful and relevant.

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**List of application already developed**

- Document Approval Portal
- Application for Project Course
- TA Allocation
- Faculty Course Load Evaluation Application
- Server Room Temp Monitoring Dashboard
- Network Printer Access Portal